

Prequalification trials

Most of the common failures (unexpected reset and program counter corruption) can be reproduced by manually forcing a low state on the NRST pin or the Oscillator pins for 1 second.

To complete these trials, ESD stress can be applied directly on the device, over the range of specification values. When unexpected behavior is detected, the software can be hardened to prevent unrecoverable errors occurring (see application note AN1015).

Electromagnetic interference (EMI)

The electromagnetic field emitted by the device are monitored while a simple application is executed (toggling 2 LEDs through the I/O ports). This emission test is compliant with IEC 61967-2 standard which specifies the test board and the pin loading.

The following table gives the EMI characteristics for f_{HSL48} and f_{HCLK} of 48 MHz.

Table 45. EMI characteristics

Symbol	Parameter	Conditions	Monitored frequency band	Max vs. $[f_{HSE}/f_{CPU}]$	Max vs. $[f_{HSL}/f_{CPU}]$	Unit
				48 MHz / 48 MHz	48 MHz / 48 MHz	
S_{EMI}	Peak ⁽¹⁾	$V_{DD} = 3.6\text{ V}$, $T_A = 25\text{ °C}$, TSSOP20 package compliant with IEC 61967-2	0.1 MHz to 30 MHz	3	3	dBμV
			30 MHz to 130 MHz	5	-2	
			130 MHz to 1 GHz	1	-1	
			1 GHz to 2 GHz	7	8	
	Level ⁽²⁾		0.1 MHz to 2 GHz	2	2	-

1. Refer to AN1709, section *EMI radiated test*

2. Refer to AN1709, section *EMI level classification*

5.3.11 Electrical sensitivity characteristics

Based on three different tests (ESD, LU) using specific measurement methods, the device is stressed in order to determine its performance in terms of electrical sensitivity.

Electrostatic discharge (ESD)

Electrostatic discharges (a positive then a negative pulse separated by 1 second) are applied to the pins of each sample according to each pin combination. The sample size depends on the number of supply pins in the device (3 parts × (n+1) supply pins). This test conforms to the ANSI/JEDEC standard.